DSA1080B: DATA SCIENCE AND ANALYTICS

INDIVIDUAL PROJECT

Global Health Data Analysis (COVID or Disease Trends)

SPRING 2026

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COVID-19 Global Data-set Analysis

Introduction

The COVID-19 pandemic has had a significant impact on countries around the world, affecting public health systems, economies, and daily life.
Analyzing data related to the spread of the virus is essential to better understand its evolution and its effects across different regions.

This report presents an exploratory and advanced analysis of COVID-19 data across multiple countries. The objective is to identify patterns, compare trends, and gain insights into how the pandemic has developed over time.

By using statistical methods, data grouping techniques, and visualizations, this analysis aims to highlight key findings and provide meaningful recommendations that can support better decision-making in managing similar health crises in the future.

Dataset Information

The COVID-19 dataset used in this analysis was obtained from *Our World in Data*, a reliable and widely recognized source of global statistics.

The dataset contains approximately 567,000 rows and multiple columns (including variables such as country, date, total cases, new cases and deaths) covering a large number of countries over a significant time period.

It is **a** time-series dataset, where each row represents daily data for a specific country, allowing for the analysis of trends and patterns over time.

The dataset includes key variables such as:

- Total cases and new cases
- Total deaths and new deaths
- Population and related indicators

Some missing values were observed, particularly in the early stages of the dataset, due to delays or inconsistencies in data reporting across countries.

Overall, the dataset provides a comprehensive view of the global evolution of COVID-19, making it suitable for both exploratory and advanced analysis.

Research Questions:

1. How has the number of confirmed COVID-19 cases changed over time in different countries?

2. Which countries have the highest mortality rates and how do they correlate with case numbers?

3. What patterns can be observed in COVID-19 testing rates across regions and over time?

4. How have vaccination rates influenced the number of new COVID-19 cases?

Dataset Size

The original dataset contained **576,000 rows**. After data cleaning and preprocessing, the final dataset contains **570,606 rows** and **61 columns**, ready for analysis.

Dataset and Repository Links

The complete project, including notebooks, code, and visualizations, is available on GitHub:

https://github.com/cizafortunette06-sketch/DSA1080A-SS2026-FortunateCiza-Global-Health-Data-Analysis-COVID-19-.git

The dataset used in this project was obtained from **Our World in Data**. You can access it here:

CSV Raw Links: catalog.ourworldindata.org/garden/covid/latest/compact/compact.csv

Data Overview

The dataset I used contains detailed information about COVID-19 cases across multiple countries, including total cases, new cases, and deaths.

It is a time-series dataset, meaning that the data is recorded daily for each country. This allows us to track how the pandemic evolved over time and observe patterns such as increases, peaks, and declines in cases.

Summary Statistics

To start the analysis, I generated summary statistics such as the mean, minimum, and maximum values.

This helped me better understand the overall distribution of the data, identify trends, and detect missing values. It also provided a quick overview of how COVID-19 cases and deaths vary across different countries.

Grouping & Analysis

After that, I grouped the data by country and calculated key metrics such as total cases and total deaths.

This made it easier to compare countries and identify those most affected by the pandemic. It also helped highlight differences in how the virus impacted different regions.

Visualizations

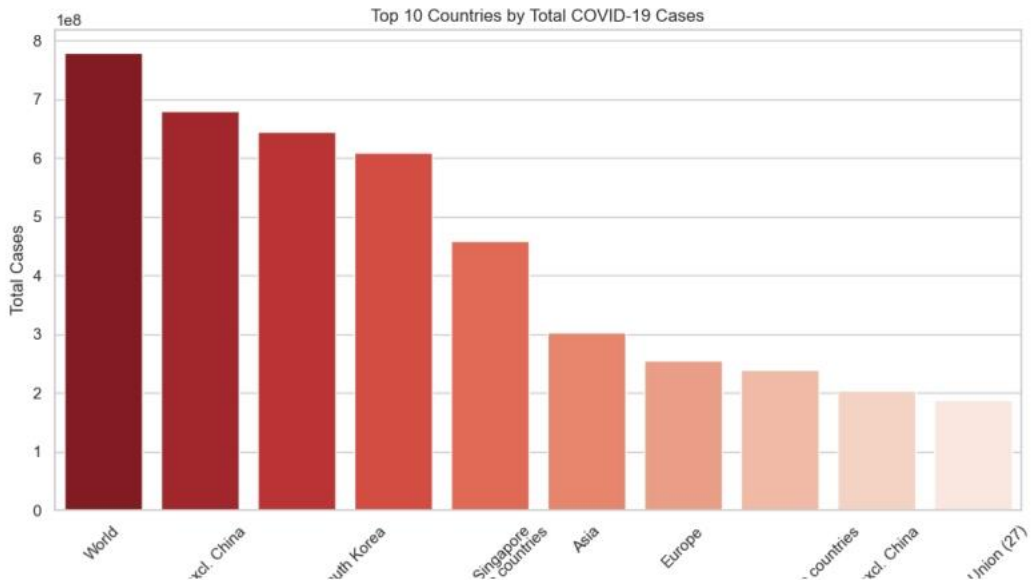
Bar Chart (Top Countries by Total Cases)

Interpretation:

The bar chart shows that a small number of countries recorded significantly higher total

COVID-19 cases compared to others.

This indicates that the impact of the pandemic was uneven across countries, with some experiencing much higher levels of infection.

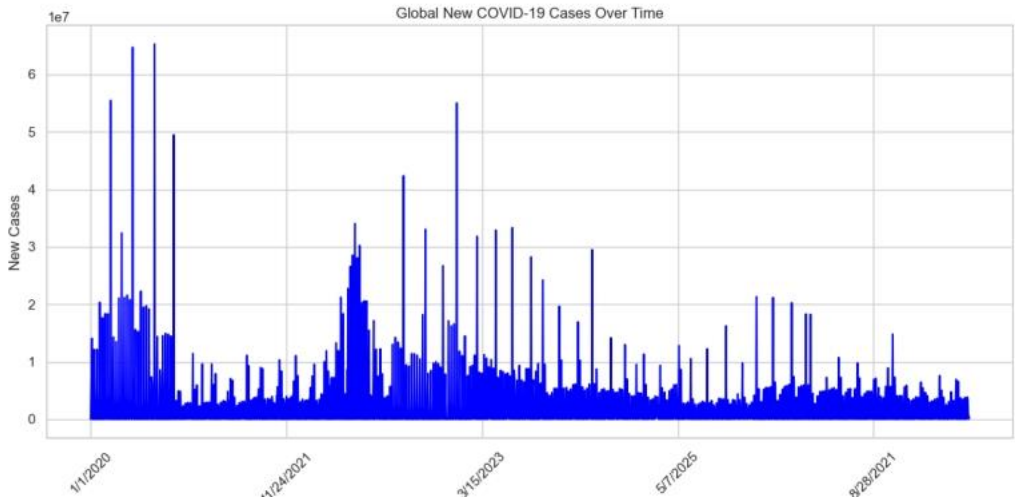


Line Chart (Global New Cases Over Time)

Interpretation:

The line chart illustrates the evolution of global COVID-19 cases over time.

Clear waves can be observed, reflecting periods of rapid increase followed by decline, which correspond to different phases of the pandemic.

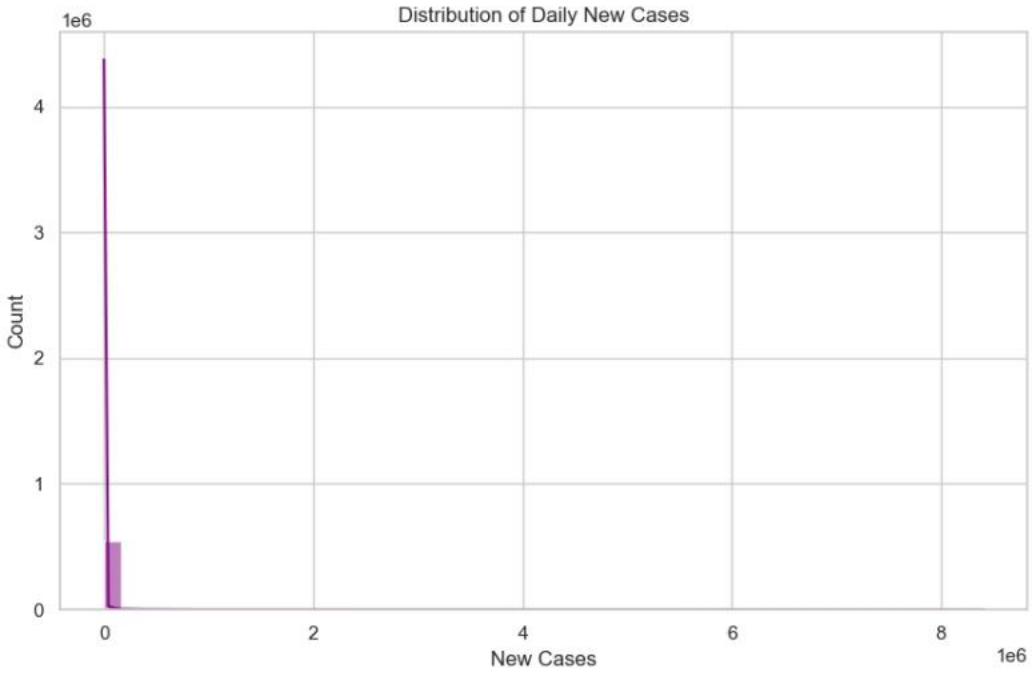


Distribution Plot (Daily New Cases)

Interpretation:

The distribution plot shows that most daily new case values are relatively low, while a smaller number of observations represent very high values.

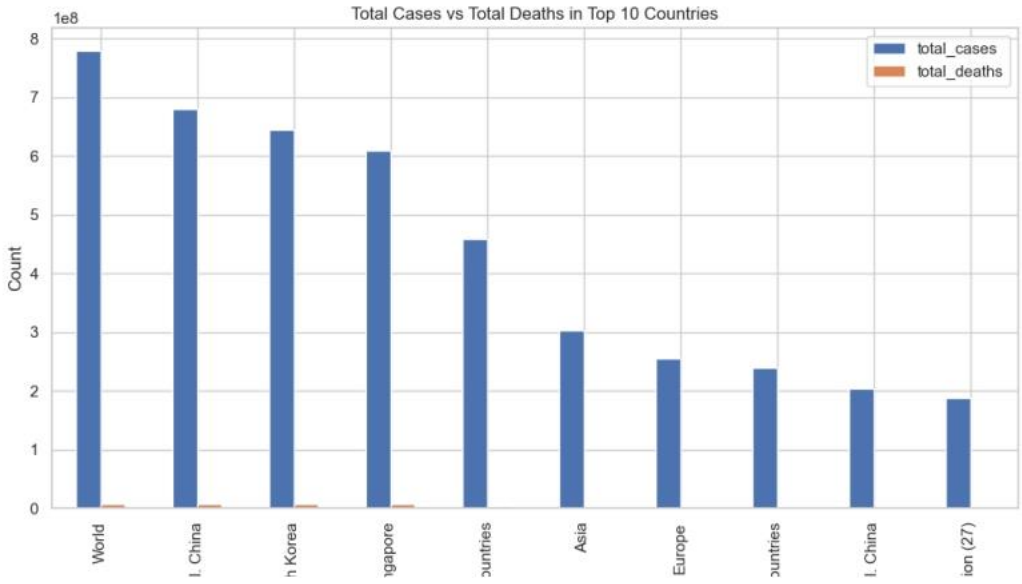
This suggests that the data is skewed, with occasional spikes indicating sudden outbreaks.



Comparative Chart (Cases vs Deaths)

Interpretation:

The comparison between total cases and total deaths shows a general positive relationship, where countries with higher case numbers also tend to have higher deaths. However, the differences across countries suggest variations in healthcare systems, response measures, and population factors.

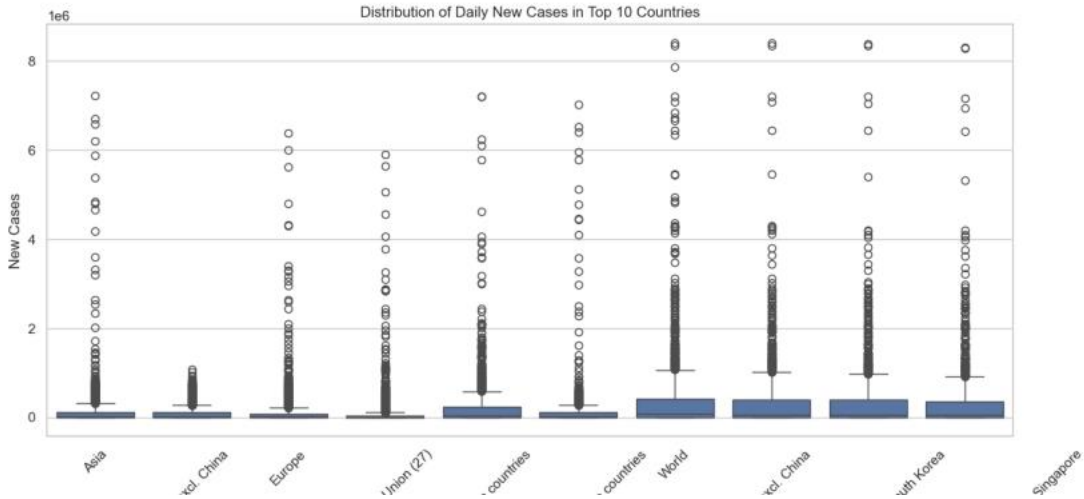


Boxplot (Variation in Daily Cases)

Interpretation:

The boxplot highlights the variability of daily new cases across countries.

Some countries show relatively stable case numbers, while others experience significant fluctuations and extreme values, indicating inconsistent patterns of infection.



Key Insights

From the analysis, the following key insights were identified:

1. **Concentration of Cases:** A small number of countries account for the majority of COVID-19 cases, highlighting that the pandemic's impact was not evenly distributed globally.
2. **Waves of Infection:** The pandemic shows clear waves over time, indicating periods of rapid spread followed by declines, which could reflect outbreaks and interventions.
3. **Discrepancy Between Cases and Deaths:** Some countries with high case numbers experienced relatively lower death rates, suggesting differences in healthcare systems, population demographics, or response strategies.
4. **Variation Across Countries:** There is significant variation in daily new cases between countries, with some experiencing extreme spikes while others remain more stable.
5. **Outbreak Patterns:** Sudden spikes in cases point to outbreaks that occurred irregularly, emphasizing the unpredictable nature of the pandemic in certain regions.

Recommendations

Based on the findings from the analysis, the following recommendations are proposed:

1. **Strengthen Early Detection Systems:** Governments should improve monitoring and early warning systems to quickly identify and respond to sudden increases in cases.
2. **Allocate Healthcare Resources Efficiently:** More medical resources and support should be directed toward countries or regions that are highly affected to better manage the impact of the pandemic.
3. **Maintain Preventive Measures During Peaks:** Preventive measures such as public awareness, testing, and health protocols should be reinforced during periods of high infection to reduce the spread of the virus.

Conclusion

This project provides an analysis of the global COVID-19 dataset from Our World in Data. It highlights key trends, variations across countries, and important insights about the pandemic's waves, case concentration, and discrepancies between cases and deaths. The visualizations and findings can help understand the global impact of COVID-19 and support further research or decision-making.